

Final Exam

Applied Optimization

Time: Feb 11, 2021 10:00 AM - 11:30 AM

Procedures:

Write your **first and last name**, put your **signature** as well as your **UniBe student number** on every page. Write down the page number and the task number. Every section must start with a new page. Use only black or blue ink pens and write legibly. Exam questions done in pencil or other colored pens will **NOT** be graded. Check if all **13 pages** are present in your exam copy.

Regulations:

Only one sheet of A4 paper with handwritten notes on both sides is permitted. Course notes, textbooks, other books/printed materials, calculators are **NOT** permitted during the examination. The cellphone and laptops are **ONLY** allowed for the exam meeting, reading the exam sheet, scanning and uploading solutions. The use of unnecessary electronic devices such as PDAs, MP3 players, iPods, etc. is strictly **PROHIBITED** during the examination.

<i>Sections</i>	<i>Points</i>	<i>Earned points</i>
1. Convex Sets	9	
2. Convex Functions	8	
3. KKT conditions	5	
4. Line Search	6	
5. Trust Region Method	6	
6. Active Set Method	12	
7. Partial Separability	6	
8. Algorithms	15	
9. Programming	6	
Total	73	

Declaration of Independence:

"I declare herewith that I wrote this examination on my own, without the help of others and without using any unpermitted sources of information (internet, books, etc.). I understand that if I do not follow these rules my examination will be assigned the mark 1.0, and that the Senate of the University of Bern is authorized to revoke the title awarded on the basis of this examination (according to Article 36, paragraph 1, litera r of the University Act of September 5th, 1996).

Write and sign the following sentence on this page: "By submitting this exam, I confirm that I have read the declaration of independence and the notes on the exam and that I accept it."

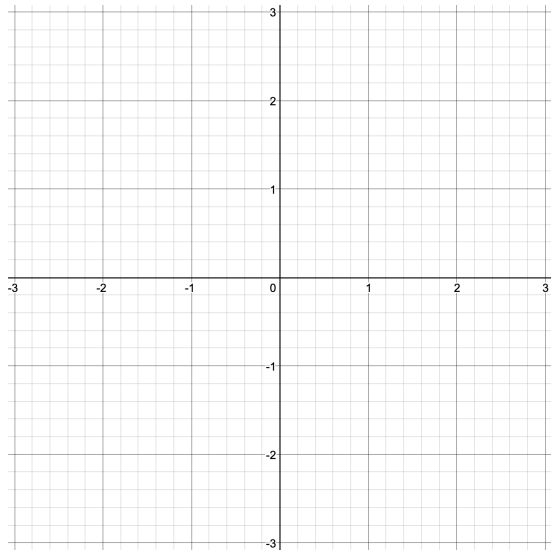
Scan this page with your student ID card on top of it.

1 Convex Sets (9 points)

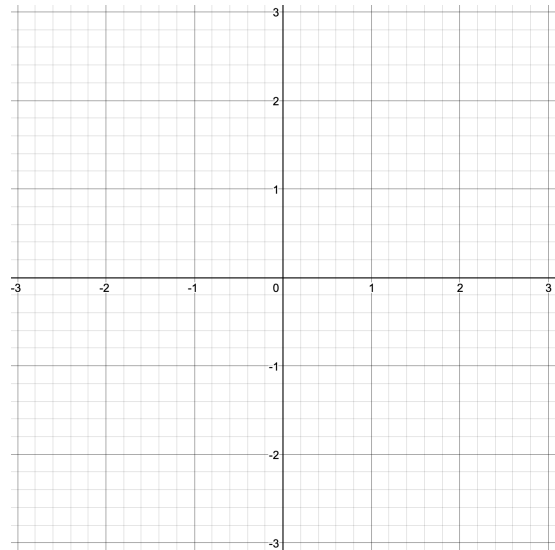
1.1. (a) Sketch the following sets. (4 points)

- $A = \{x \in \mathbb{R} \mid \|x\|_1 \leq 1\}$
- $B = \{(x, y) \in \mathbb{R}^2 \mid \|x\|_2 \leq y, y \leq 2\}$
- $C = \{x \in \mathbb{R}^2 \mid \|x\|_1 \geq 3, \|x\|_\infty \leq 2\}$
- $D = \{x \in \mathbb{R}^2 \mid \|x\|_1 \leq 1, x \neq (1, 0)\}$

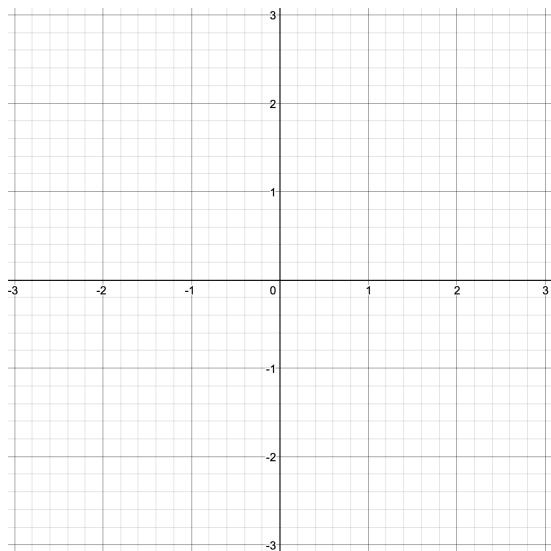
(b) State which of the above sets are convex: _____. (1 point)



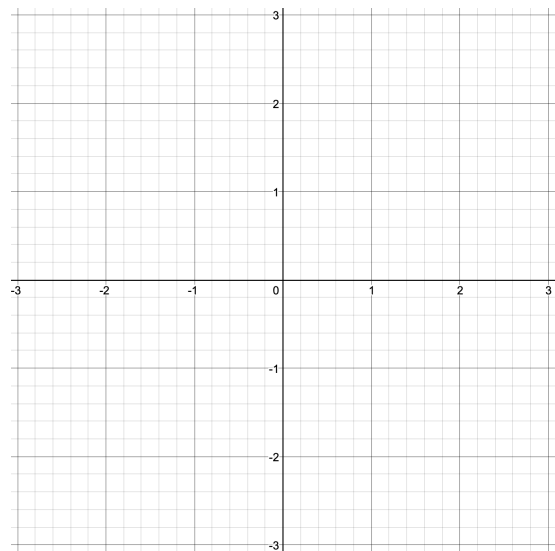
A



B



C



D

1.2. Prove that the unit norm ball $B = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$ is a convex set. (4 points)

Hint: A norm $\|\cdot\|$ satisfies the following properties:

(1 - non-negativity) $\|x\| \geq 0$

(2 - definiteness) $\|x\| = 0 \Leftrightarrow x = 0$

(3 - homogeneity) $\|t \cdot x\| = |t| \|x\|$ for $t \in \mathbb{R}$

(4 - triangle inequality) $\|x + y\| \leq \|x\| + \|y\|$

2 Convex Functions (4+4 points)

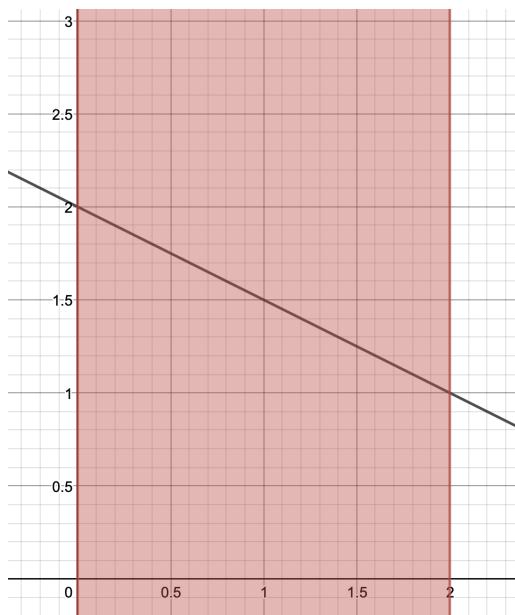
2.1. Let $f(x)$ and $g(x)$ be two convex functions with $x \in \mathbb{R}$. Is $h(x) = f(x) \cdot g(x)$ always convex? Prove convexity or give a counter example (and show that it is non-convex).

2.2. Let $f(x)$ and $g(x)$ be two convex functions with $x \in \mathbb{R}$. Is $h(x) = f(g(x))$ always convex? Prove convexity or give a counter example (and show that it is non-convex).

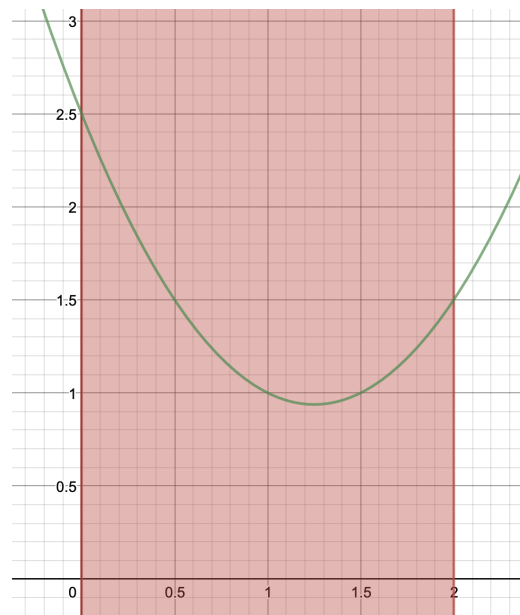
3 KKT conditions (5 points)

In the figure below, four different functions (f_1) – (f_4) are plotted with the constraints $0 \leq x \leq 2$.

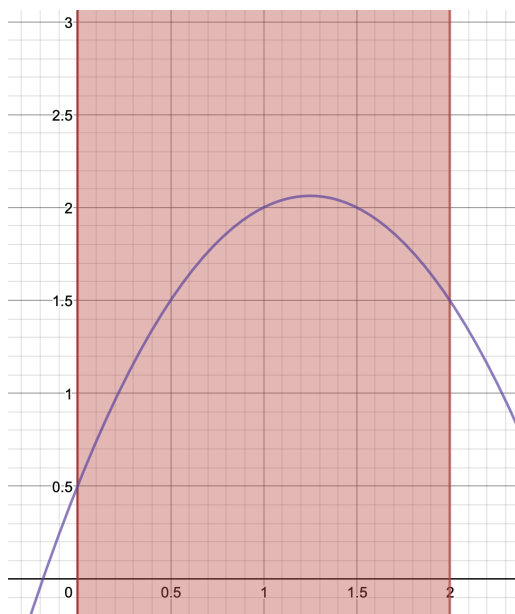
- (a) Identify points in the graphs where the KKT conditions can be satisfied (with respect to minimization)? Mark them with $\circ$ in the figures. (4 points)
- (b) Which points are global optima of the constrained optimization problems? Mark the points with $\times$ in the figures. (1 point)



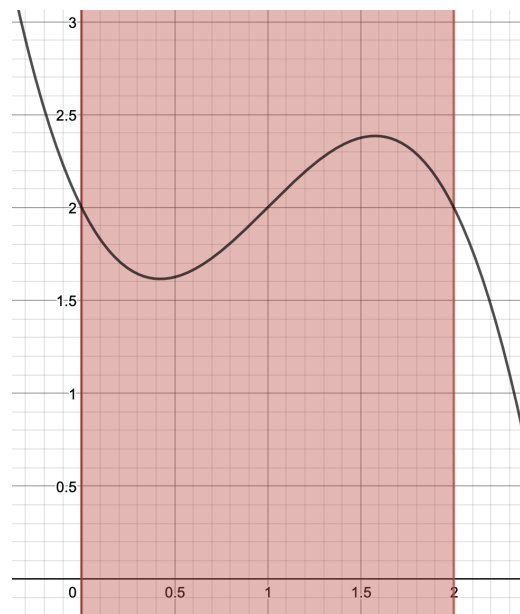
(f_1)



(f_2)



(f_3)



(f_4)

4 Line Search (6 points)

Consider the objective function

$$f(x,y) = (x-1)^2 + \frac{1}{4}(y-1)^2$$

- (a) Given $p = (0, 1)^T$ and $v = (2, 0)^T$ write the expression for $f(x,y)$ restricted to the line $p + tv$ and compute t^* . (2 points)
- (b) What is the interval for t that a backtracking line search satisfying the Armijo condition with $\alpha = 1/8$ would consider? Explain why and provide the computations. (2 points)
- (c) What is the interval for t that a backtracking line search satisfying the Wolfe conditions with $\alpha = 1/8$ and $\beta = 1/4$ would consider? Explain why and provide the computations. (2 points)

5 Trust Region Method (6 points)

Which is in general true about trust region methods. Mark true or false in each case.

(1 points for marking a true statement, -1 points for marking a false statement; negative totals are clamped to 0)

True	False	Question
		Trust region methods compute first a search direction v like line search methods and then a step length λ such that the solution stays inside the trust region $\ \lambda v\ ^2 - \Delta^2 \leq 0.$
		The size of the trust region Δ is updated with heuristics based on local information about the accuracy of the model $m(v)$ with respect to the objective function $f(x)$.
		The solution of the trust region subproblem lies always at the boundary of the trust region.
		At a given iteration of the trust region subproblem, the solution is at the boundary of the trust region. This means that the radius Δ was too small and needs to be increased.
		If on a given iteration the solution of the subproblem $x_{k+1} = x_k + v_k$ makes the value of $f(x_k) - f(x_{k+1})$ close to $m_k(0) - m_k(v_k)$, then the trust region radius Δ_k will be decreased for next iteration.
		The Dogleg method requires the coefficient matrix B of the quadratic model to be positive definite, while the Cauchy Point method doesn't.

6 Active Set Method (12 points)

6.1. Consider the following equality constrained optimization problem

$$\begin{array}{ll}\text{minimize} & \frac{1}{2}x^T Hx \\ \text{subject to:} & Ax = b\end{array}$$

with $H \in \mathbb{R}^{n \times n}$ being symmetric positive definite, $A \in \mathbb{R}^{n \times n}$ and $x \in \mathbb{R}^n$.

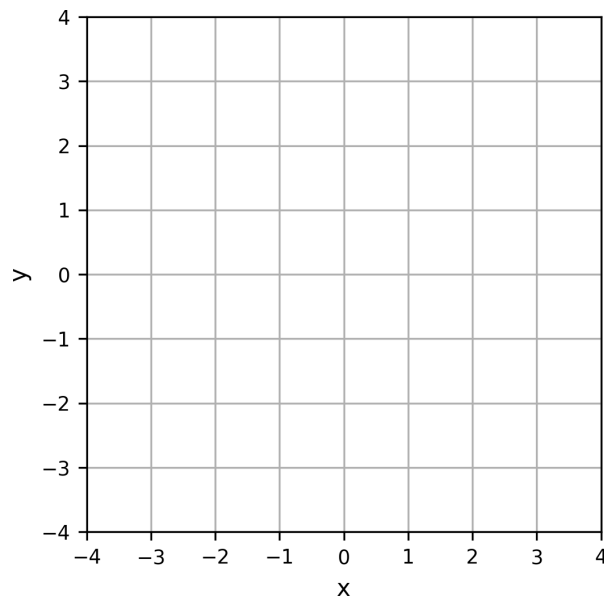
(a) Derive the KKT conditions. (3 points)

(b) State the KKT conditions as a symmetric system of linear equations. (1 point)

6.2. Consider the following optimization problem with $x, y \in \mathbb{R}$

$$\begin{array}{ll}\text{minimize} & \frac{1}{2}(x^2 + y^2) \\ \text{subject to:} & x \geq 1 \quad (C1) \\ & y \geq -2 \quad (C2) \\ & -x \geq -3 \quad (C3) \\ & -y \geq -2 \quad (C4)\end{array}$$

(a) Sketch the feasible set in the grid below. (1 point)



(b) Sketch the path of the active set method starting from $(x_0, y_0) = (3, 1)$ with initial active set $A_0 = \{3\}$. (1 point)

(c) Verify the path of the Active Set Method computationally. (6 points)

Note for each step the following states in the table below:

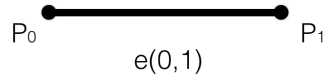
- the primal variables before the step (x_k, y_k)
- the active set before the step A_k
- the primal variables after the step (x_{k+1}, y_{k+1})
- the dual variables after the step λ_{k+1} , e.g. $(0, 1, -1, 0)$
- the active set after the step A_{k+1}
- whether the KKT conditions are satisfied for $(x_{k+1}, y_{k+1}, \lambda_{k+1})$

k	(x_k, y_k)	A_k	(x_{k+1}, y_{k+1})	λ_{k+1}	A_{k+1}	KKT sat.?
0	(3,1)	$\{3\}$				
1						
2						
3						

Hint: Solve the equality constrained subproblem with the result of problem 6.1 (b)

7 Partial Separability (6 points)

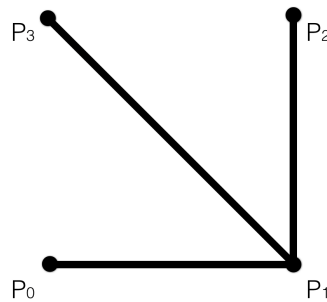
Consider the function $E_{(i,j)}(x) = \frac{1}{2} \|x_i - x_j\|_2^2$ defined on an edge $e(i,j)$, where $x_i \in \mathbb{R}$ is a function value defined on node P_i and $x \in \mathbb{R}^n$ with n being the number of nodes. Assume we have only one edge $e(0,1)$ with two nodes P_0, P_1 , as shown in the figure below.



(a) Write down the gradient of the function $E_{(0,1)}(x)$. (1 point)

(b) Write down the Hessian matrix of the function $E_{(0,1)}(x)$. (1 point)

(c) Assume we have the graph as shown in the following figure, with 4 nodes and 3 edges. The function defined on the graph is then $\hat{E}(x) = \sum_{e(i,j)} E_{(i,j)}(x)$, where $e(i,j)$ is in the edge set $\{(0,1), (1,2), (1,3)\}$. Compute the Hessian matrix $\hat{H}$ of the function $\hat{E}(x)$ as the sum of Hessian matrices of the individual edge terms. Use the template provided below to note your result. (No need to write down the entries with value 0). (4 points)



$$\begin{matrix} \hat{H}_{(0,1)} & + & \hat{H}_{(1,2)} & + & \hat{H}_{(1,3)} & = & \hat{H} \\ \begin{pmatrix} - & - & - & - \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{pmatrix} & + & \begin{pmatrix} - & - & - & - \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{pmatrix} & + & \begin{pmatrix} - & - & - & - \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{pmatrix} & = & \begin{pmatrix} - & - & - & - \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{pmatrix} \end{matrix}$$

8 Algorithms (15 points)

Consider the task of optimizing a **large-scale** problem with $n = 10^6$ variables, given that state-of-the-art implementations of the following methods are available:

- (a) Gradient Descent method
- (b) Newton's method
- (c) Gauss-Newton method
- (d) L-BFGS method
- (e) Active Set Method using Newton's solver
- (f) Primal-Dual Interior Point method using Newton's solver with Hessian modification

For each of the following scenarios, list algorithms that can find a local minimum and specify the algorithm for which you expect the shortest runtime (assuming infinite memory).

Briefly justify your choice.

- 8.1. A problem with linear inequality constraints and the Hessian matrix of the objective function is sparse. (3 points)
- 8.2. An unconstrained problem with a non-convex objective function, and the Hessian matrix is dense. (3 points)
- 8.3. A least-squares problem of the form $f(x) = \sum_{i=1}^m (g_i(x_{a(i)}, x_{b(i)}))^2$ with $g_i : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $a(i), b(i) \in \{1, \dots, n\}$. In addition, $f(x)$ is strictly convex, infinitely often differentiable, and $m = 2n$. (3 points)
- 8.4. An unconstrained problem of which only first-order derivatives exist. (3 points)
- 8.5. An unconstrained problem of which the Hessian matrix has been proven to be identity. (3 points)

9 Programming (6 points)

Assuming that we use the log barrier method to solve a sparse optimization problem, the following code evaluates the Hessian of the problem. The variables with underscore in the end are global variables that are declared.

```
1 // sum up hessian matrices (objective + barrier functions)
2 using SMat = Eigen::SparseMatrix<double>; //sparse
3 using Vec = Eigen::VectorXd; // dense
4
5 virtual void eval_hessian(const Vec &_x, SMat &_H) override {
6     //get hessian of the original objective and it's sparse
7     obj_>eval_hessian(_x, _H);
8     _H *= t_;
9
10    //add hessian of constraints
11    int n = obj_>n_unknowns();
12    Vec g(n);
13    SMat M(n, n);
14    for (unsigned int i = 0; i < constraints_.size(); ++i) {
15        // get f, grad, hess
16        double d = constraints_[i]>eval_constraint(_x);
17        g.setZero();
18        constraints_[i]>eval_gradient(_x, g);
19        M.setZero();
20
21        //get hessian of the barrier functions and they're sparse
22        constraints_[i]>eval_hessian(_x, M);
23
24        //assemble hessian matrices
25        _H += (1.0 / (d * d)) * g*g.transpose() - (1.0 / d) * M;
26    }
27 }
```

(a) Mark the most inefficient line of the code and describe the problem. (2 points)

(b) Describe how to significantly improve performance. (4 points)